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Implementation of Argo Real Time Quality Controls by Coriolis

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Implementation of Argo Real Time Quality Controls by Coriolis

Authors: Jean-Philippe Rannou, Cécile Cabanes, Christine Coatanoan, Catherine Lagadec and Virginie Thierry

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History

Date (dd/mm/yyyy)	Comment
09/02/2016	Creation of the document by Cecile, Jean-Philippe
18/042017	Update of the draft document to create its first version, Jean-Philippe
17/05/2019	Modification of pressure threshold determination for test #19
20/09/2019	Modification of TEMP (TEMP_DOXY) specific range test in the Red Sea. Added global range test for DOWN_IRRADIANCE380, DOWN_IRRADIANCE412, DOWN_IRRADIANCE443, DOWN_IRRADIANCE490 and DOWNWELLING_PAR parameters.
05/03/2020	Updated according to version 4.5 of Coriolis RTQC code (updated to be compliant with version 3.3 of "Argo Quality Control Manual for CTD and Trajectory Data").
19/06/2020	Updated according to version 4.6 (profiles) and 2.9 (trajectory) of Coriolis RTQC code.
05/04/2022	Updated according to version 5.8 (profiles) and 3.4 (trajectory) of Coriolis RTQC code (to cope with version 3.6 of Argo Quality Control Manual for CTD and Trajectory Data).
03/03/2025	Updated according to version 7.3 (profiles) and 4.2 (trajectory) of Coriolis RTQC code (to cope with version 3.9 of Argo Quality Control Manual for CTD and Trajectory Data).

Reference Documents

Reference N°	Title	Link
#RD1	Argo Quality Control Manual for CTD and Trajectory Data	http://dx.doi.org/10.13155/33951
#RD2	Argo quality control manual for biogeochemical data	http://dx.doi.org/10.13155/40879
#RD3	Argo quality control manual for Dissolved Oxygen Concentration	http://dx.doi.org/10.13155/46542
#RD4	BGC Argo quality control manual for particles backscattering	https://doi.org/10.13155/60262
#RD5	Bio-Argo quality control manual for Chlorophyll-A concentration	http://dx.doi.org/10.13155/35385
#RD6	BGC-Argo quality control manual for nitrate concentration	https://doi.org/10.13155/84370
#RD7	BGC-Argo quality control for radiometry	https://doi.org/10.13155/62466
#RD8	BGC-Argo quality control manual for pH	https://doi.org/10.13155/97828

1. Introduction

This document describes how the Real Time Quality Control (RTQC) tests are implemented at the Coriolis data Centre.

These tests are defined in the Argo quality control manuals (#RD1 and #RD2). The present document provides a detailed description of their implementation in the Matlab processing chain that produces Argo V3.1 format NetCDF files.

This document describes the tests implemented in **profile RTQC version 7.3** and **trajectory RTQC version 4.2** of the Coriolis **Matlab decoder version '074a'** (http://doi.org/10.17882/45589).

The implemented tests are defined for the following parameters:

- PRES ,
- TEMP,
- PSAL,
- CNDC,
- TEMP_CNDC,
- DOXY,
- TEMP_DOXY,
- CHLA,
- CHLA_FLUORESCENCE,
- BBP700,
- PH_IN_SITU_TOTAL,
- NITRATE,
- DOWN_IRRADIANCE380,
- DOWN_IRRADIANCE412,
- DOWN_IRRADIANCE443,
- DOWN_IRRADIANCE490,
- DOWN_IRRADIANCE555,
- DOWN_IRRADIANCE665,
- DOWN_IRRADIANCE670,
- DOWNWELLING_PAR.

2. Implementation of Argo Real-time Quality Control test procedures on vertical profiles

2.1. Description of implemented tests

2.1.1. Test #1: Platform identification test

The Coriolis Matlab decoder cannot process a float without its WMO and associated PTT. Thus, nothing has been implemented for this test (it is always reported as successfully passed).

2.1.2. Test #2: Impossible date test

At Coriolis the JULD value is checked to be later than January 1st 1997 (JULD ≥ 17167) and earlier than the date of the check (in UTC time).

The JULD_LOCATION is also checked with the same criteria and contributes (with test #3, #4 and #5) to the final value of POSITION_QC.

2.1.3. Test #3: Impossible location test

At Coriolis the checked interval is [-90, 90] for latitudes and [-180, 180] for longitudes.

2.1.4. Test #4: Position on land test

At Coriolis we use the GEBCO_2024 file (https://www.bodc.ac.uk/data/open_download/gebco/gebco_2024/zip/).

The detailed description of the test is the following: For a given position, the elevations are retrieved from the bathymetry file (1, 2 or 4 elevations depending on the position location on the GEBCO grid). The test fails if the average value of the retrieved elevations is ≥ 0.

2.1.5. Test #5: Impossible speed test

At Coriolis, the profile position is defined according to DAC cookbook specifications (§2.2.2 of http://dx.doi.org/10.13155/29824), thus from a trajectory already checked by the JAMSTEC QC test. Consequently, this test is obsolete for Coriolis Argos floats and should be considered for Iridium floats only.

2.1.6. Test #6: Global range test

At Coriolis the checked intervals are:

For the PRES parameter:

- PRES < -5 dbar to flag ‘4’ PRES, TEMP and PSAL,
- -5dbar ≤ PRES ≤ -2.4 dbar to flag ‘3’ PRES, TEMP and PSAL.

For other parameters:

- TEMP in the interval [-2.5, 40] °C,
- PSAL in the interval [2, 41] PSU,
- DOXY in the interval [−5, 600] micromoles/kg,
- TEMP_DOXY in the interval [-2.5, 40] °C,

- CHLA in the interval [−0.2, 100] mg/m^3,
- CHLA_FLUORESCENCE in the interval [−0.2, 100] ru,
- PH_IN_SITU_TOTAL in the interval [7.0, 8.8],
- PH_IN_SITU_TOTAL_ADJUSTED in the interval [7.3, 8.5],
- NITRATE in the interval [-2, 50] micromoles/kg,
- DOWN_IRRADIANCE380 in the interval [-1, 1.7] W/m^2/nm,
- DOWN_IRRADIANCE412 in the interval [-1, 2.9] W/m^2/nm,
- DOWN_IRRADIANCE443 in the interval [-1, 3.2] W/m^2/nm,
- DOWN_IRRADIANCE490 in the interval [-1, 3.4] W/m^2/nm,
- DOWN_IRRADIANCE665 in the interval [-1, 2.8] W/m^2/nm,
- DOWNWELLING_PAR in the interval [-1, 4672] microMoleQuanta/m^2/sec.

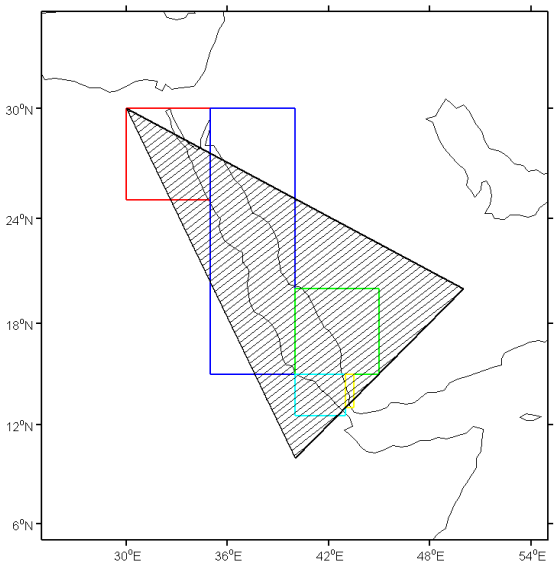
Values that fail this test are flagged ‘4’.

2.1.7. Test #7: Regional range test

At Coriolis the regions are defined by rectangular areas.

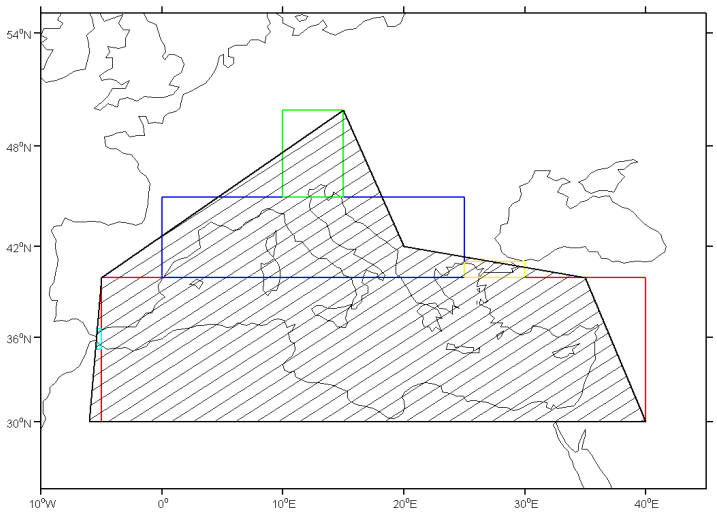
For the Red Sea Region, 5 rectangular areas:

- 30N, 30E ; 25N, 35E,
- 30N, 35E ; 15N, 40E,
- 20N, 40E ; 15N, 45E,
- 15N, 40E ; 12.55N, 43E,
- 15N, 43E ; 13N, 43.5E.



For the Mediterranean Sea region, 5 rectangular areas:

- 40N, 5W ; 30N, 40E,
- 45N, 0; 40N, 25E,
- 50N, 10E; 45N, 15E,
- 41N, 25E; 40N, 30E,
- 36.6N, 5.4W; 35.2N, 5W.



The borders are considered as part of the region.

The checked intervals are:

- TEMP and TEMP_DOXY in the interval [21, 40] °C in the Red Sea and [10, 40] °C in the Mediterranean Sea,
- PSAL in the interval [2, 41] PSU in the Red Sea and [2, 40] PSU in the Mediterranean Sea.

2.1.8. Test #8: Pressure increasing test

Some BGC floats have the ability to transmit 'raw' sampled data, i.e. direct sensor output without any averaging or decimation of the measurements. Some parts of these 'raw' sampled profiles will probably be flagged as bad by this test (at the beginning and end parts of the profile).

As shown in the table below, the flagged data depend on the implementation (if we start the check from shallow values, from deep values or from the middle of the profile).

PRES	QC (start shallow)	QC (start deep)	QC (start middle)
0	1	1	1
1	1	4	4
2	1	4	4
1	4	1	1
2	4	4	4
3	1	4	4
4	1	4	4
3	4	4	4
2	4	1	1
3	4	4	4
3	4	1	1
4	4	1	1
5	1	1	1

6	1	4	4
7	1	4	4
8	1	4	4
9	1	4	4
10	1	4	4
9	4	4	4
8	4	4	4
7	4	4	4
6	4	1	1
7	4	1	1
8	4	1	1
9	4	1	1
10	4	1	1
11	1	1	1
12	1	1	1
13	1	1	1
14	1	1	1
15	1	1	1
…	…	…	…
1995	1	1	1
1996	1	1	1
1997	1	4	1
1998	1	4	1
1999	1	4	1
2000	1	4	1
2001	1	4	1
2002	1	4	1
2002	4	4	4
2001	4	4	4
2000	4	4	4
2000	4	4	4
1999	4	4	4
1998	4	4	4
1997	4	1	4

At Coriolis:

- For near-surface profiles, we start from the deepest measurement of the profile and check increasing pressures,
- For other profiles, we start from the middle of the profile and check increasing pressures to deep measurements and decreasing pressures to shallow measurements.

2.1.9. Test #9: Spike test

This test is implemented for TEMP, PSAL, DOXY, TEMP_DOXY, CHLA, PH_IN_SITU_TOTAL and NITRATE according to Argo quality control manuals (#RD1 and #RD2).

It is not implemented yet for BBP700 and BBP532 (Catherine SCHMECHTIG's decision).

2.1.10. Test #10: Top and bottom spike test: obsolete

2.1.11. Test #11: Gradient test

This test is implemented for DOXY according to Argo quality control manual (#RD2).

2.1.12. Test #12: Digit rollover test

The test provides thresholds to detect a jump between two adjacent measurements of a parameter. It then specifies that the values that failed the test should be flagged as bad data. However, when a jump is detected, we cannot be sure in real time which part of the profile has rollover values

At Coriolis, when a jump is detected in a measured parameter (with the specified criteria) the 'jumped values' are flagged '4' and all the remaining values of the profile for this parameter are flagged '3'.

This test is implemented for TEMP, PSAL and TEMP_DOXY.

2.1.13. Test #13: Stuck value test

According to Argo quality control manuals (#RD1 and #RD2) this test is implemented for all 'c' and 'b' parameters and their associated 'intermediate' parameters.

2.1.14. Test #14: Density inversion

This test has been implemented thanks to Cécile Cabanes's additional Matlab implementation example provided in Annex A of the present document.

2.1.15. Test #15: Supplemental sensor exclusion list test

This test is implemented according to Argo quality control manuals (#RD1 and #RD2).

2.1.16. Test #16: Gross salinity or temperature sensor drift

In the Coriolis implementation of this test we search the previous good profile within the multi-profile file (thus from primary sampled profiles only) and within the profiles of the same direction as the checked profile.

This test is implemented for TEMP, PSAL and TEMP_DOXY.

However, this test is designed for floats that always profile from the same pressure and not for:

- Argos floats with multi-mission,
- Iridium floats with a mission that can be modified at sea.

Consequently, we finally decided to not apply this test in Coriolis RTQC.

2.1.17. Test #17: Visual QC

This test is performed by Coriolis operators.

2.1.18. Test #18: Frozen profile test

This test is implemented for TEMP, PSAL and TEMP_DOXY.

However, this test is designed for floats with constant cycle duration of ~10 days and not for:

- Argos float versions with different programmed cycle durations,
- Iridium floats with a mission that can be modified at sea.

Consequently, we finally decided to not apply this test in Coriolis RTQC.

2.1.19. Test #19: Deepest pressure test

This test is implemented according to version 3.9 of Argo quality control manual (#RD1).

Note however that when PRES ≤ PRESSURE_THRESHOLD, this test will flag all profile parameters as good data. This is not acceptable for parameters that have no other RTQC test already defined.

Consequently, we apply this test to all parameters without initializing QC data to '1'.

2.1.20. Test #25: MEDian with a Distance (MEDD) test

This test is implemented on (PRES, TEMP, PSAL), (PRES, TEMP) and (PRES, TEMP_DOXY) profiles.

2.1.21. Test #57: DOXY specific test

These tests are implemented according to the dedicated Argo quality control manual (#RD3).

2.1.22. Test #58: CDOM specific test

No test specific to CDOM parameter is implemented yet.

2.1.23. Test #59: NITRATE specific test

These tests are implemented according to the dedicated Argo quality control manual (#RD6).

2.1.24. Test #60: PAR specific test

No test specific to PAR parameter is implemented yet.

2.1.25. Test #61: IRRADIANCE specific test

No test specific to IRRADIANCE parameter is implemented yet.

2.1.26. Test #62: BBP specific test

No test specific to BBP parameter is implemented yet.

The “bad offset test” will be implemented as soon as the dedicated Argo quality control manual (#RD4) is available.

2.1.27. Test #63: CHLA specific test

These tests are implemented according to the dedicated Argo quality control manual (#RD5).

2.2. Tests application order on vertical profiles

At Coriolis, Argo real-time QC tests on vertical profiles are applied in the order described in the following table.

Order	Test number	Test name
1	1	Platform identification
2	2	Impossible date test
3	3	Impossible location test
4	4	Position on land test
5	5	Impossible speed test
6	15	Supplemental sensor exclusion list test
7	19	Deepest pressure test
8	21	Near-surface unpumped CTD salinity test
9	22	Near-surface mixed air/water test
10	6	Global range test
11	7	Regional range test
12	8	Pressure increasing test
13	9	Spike test
14	11	Gradient test
15	25	MEDD test
16	12	Digit rollover test
17	13	Stuck value test
18	14	Density inversion test
not used	16	Gross salinity or temperature sensor drift test
not used	18	Frozen profile test
19	26	TEMP_CNDC test
20	24	Pre-april 2021 RBR floats
21	23	Real-time Quality Control flag scheme for float data deeper than 2000 dbar
-	27 to 55	Not used
22	56	PH specific test
23	57	DOXY specific test
not defined yet	58	CDOM specific test
24	59	NITRATE specific test
25	60	PAR specific test
26	61	IRRADIANCE specific test
27	62	BBP specific tests
28	63	CHLA specific tests

3. Implementation of Argo Real-time Quality Control test procedures on near-surface data of vertical profiles

3.1. Description of implemented tests

All tests, listed in § 2.1 are implemented and applied on near-surface data of vertical profiles. Note that for "Test #8: Pressure increasing test", the starting point is the deepest measurement and the check is performed on increasing pressures.

Two additional specific tests have been implemented for near-surface data of vertical profiles.

3.1.1. Test #21: Near-surface unpumped CTD salinity test

This test is implemented for PSAL and DOXY according to Argo quality control manuals (#RD1 and #RD2).

3.1.2. Test #22: Near-surface mixed air/water test

This test is implemented for TEMP and TEMP_DOXY according to Argo quality control manuals (#RD1 and #RD2).

4. Implementation of Argo Real-time Quality Control test procedures on trajectories

4.1. Description of implemented tests

4.1.1. Test #1: Platform identification test

See § 2.1.1.

4.1.2. Test #2: Impossible date test

At Coriolis the JULD value is checked to be later than January 1st 1997 (JULD ≥ 17167) and earlier than the date of the check (in UTC time).

4.1.3. Test #3: Impossible location test

See §2.1.3.

4.1.4. Test #4: Position on land test

See §2.1.4.

4.1.5. Test #5: Impossible speed test

This test is not implemented at Coriolis; it has been replaced by test #20: Questionable Argos position test.

4.1.6. Test #6: Global range test

See §2.1.6.

4.1.7. Test #7: Regional range test

See §2.1.7.

4.1.8. Test #15: Supplemental sensor exclusion list test

This test is implemented according to Argo quality control manuals (#RD1 and #RD2).

4.1.9. Test #20: Questionable Argos position test

This test has been implemented according to Annex H of http://dx.doi.org/10.13155/29824.

4.1.10. Test #57: DOXY specific test

These tests are implemented according to the dedicated Argo quality control manual (#RD3).

4.1.11. Test #58: CDOM specific test

No test specific to CDOM parameter is implemented yet.

4.1.12. Test #59: NITRATE specific test

The specified test is relevant for vertical profiles only.

4.1.13. Test #60: PAR specific test

No test specific to PAR parameter is implemented yet.

4.1.14. Test #61: IRRADIANCE specific test

No test specific to IRRADIANCE parameter is implemented yet.

4.1.15. Test #62: BBP specific test

See §2.1.26.

4.1.16. Test #63: CHLA specific test

The specified test is relevant for vertical profiles only.

4.2. Tests application order on trajectories

At Coriolis, Argo real-time QC tests on trajectories are applied in the order described in the following table.

Order	Test number	Test name
1	1	Platform identification
2	2	Impossible date test
3	3	Impossible location test
4	4	Position on land test
5	20	Questionable Argos position test
6	15	Supplemental sensor exclusion list test
7	21	Near-surface unpumped CTD salinity test
8	22	Near-surface mixed air/water test
9	6	Global range test
10	7	Regional range test
11	56	PH specific test
12	57	DOXY specific test
not defined yet	58	CDOM specific test
no relevant for trajectories	59	NITRATE specific test
13	60	PAR specific test
14	61	IRRADIANCE specific test
15	62	BBP specific tests
no relevant for trajectories	63	CHLA specific tests

4.3. Profile measurements reported in trajectories

The following profile measurements are duplicated in the trajectories:

- The deepest measurement of the descending and ascending profiles (duplicated and stored with MC 203 and MC 503 respectively),

- The measurements of the dated levels of the profiles (duplicated, with their JULD, and stored with MC 190 (for descending profiles) or MC 590 (for ascending profiles).

These profile measurements are duplicated with their QCs set by RTQC tests on vertical profiles.

5. Implementation of Argo Real-time Quality Control test procedures on in-air data of trajectories

Some Coriolis floats are able to sample in-air measurements which are stored in the trajectories with MC 1090.

5.1. Description of implemented tests

All tests, listed in § 4.1 are implemented and applied on in-air data of trajectories.

Two additional specific tests have been implemented for in-air data of trajectories.

5.1.1. Test #21: Near-surface unpumped CTD salinity test

See §3.1.1.

5.1.2. Test #22: Near-surface mixed air/water test

See §3.1.2.

6. Passed/failed test traceability

Passed and failed test number lists are reported in the profile and trajectory files through the HISTORY_QCTEST variable.

At Coriolis:

- All tests but tests #11, 56, 57, 59, 60, 61, 62 and 63 are reported in the C files.
- All tests but tests #8, 14, 24, 25 and 26 are reported in the B files.

7. Annex A: Example of Matlab implementation of the density inversion test

```
% [delta_upbot,delta_botup,is_inv_dens] = test_density14(P,T,S,threshold)
% INPUT:
% P(1,nlevel) pression (dB)
% T(1,nlevel) T in situ (°C)
% S(1,nlevel) S in situ (PSU)
% treshold : treshold used for detecting density inversion (0.03kg/m3)
% OUTPUT:
% delta_upbot (1,nlevel) : top to bottom test :  if delta_upbot >= 0.03kg/m3 flag 4 of the
level i
% delta_botup (1,nlevel) : bottom to top test:  if delta_botup <= -0.03kg/m3 flag of the level
i
% is_inv_dens (1,nlevel) : is_inv_dens=1 if level i should be flagued as bad (flag 4) ; ==0
otherwise

% You will need The ITS-90 version of the CSIRO SEAWATER library, which you can obtain from
% http://www.cmar.csiro.au/datacentre/ext_docs/seawater.htm.  Version 3.3   22-Sep-2010
%

function [delta_upbot,delta_botup,is_inv_dens] = test_density14(P,T,S,treshold)

if size(P,1)> 1
P=P';
end
if size(T,1)> 1
T=T';
end
if size(S,1)> 1
S=S';
end


%%
% From top to bottom :  two consecutive levels, i and i+1, are checked (Pi < Pi+1 )
%%
Pref = (P(1:end-1)+P(2:end))/2;


% rho_i(Pi, Ti, Si, Pref): potential density referenced to the mid-point Pref

rhop_i = sw_pden(S(1:end-1),T(1:end-1),P(1:end-1),Pref);

% rho_ip1(Pi+1, Ti+1, Si+1, Pref): potential density referenced to the mid-point Pref

rhop_ip1 = sw_pden(S(2:end),T(2:end),P(2:end),Pref);

delta_upbot = [rhop_i-rhop_ip1, NaN];


%%
% From bottom to top:  two consecutive levels, i and i+1, are checked (Pi > Pi+1 )
%%
Pref = (P(2:end)+P(1:end-1))/2;


% rho_i(Pi, Ti, Si, Pref): potential density referenced to the mid-point Pref

rhop_i = sw_pden(S(2:end),T(2:end),P(2:end),Pref);

% rho_ip1(Pi+1, Ti+1, Si+1, Pref): potential density referenced to the mid-point Pref

rhop_ip1 = sw_pden(S(1:end-1),T(1:end-1),P(1:end-1),Pref);

delta_botup = [NaN, rhop_i-rhop_ip1];


% test if there is density inversion with the 0.03kgm-3 criterion

is_inv_dens = delta_upbot>=treshold | delta_botup<=-treshold;
```